

DOINGG@UNION.EDU

DOING-LAB-NOTEBOOK

SELF

Contents

Home 3

Announcements 3

Something else 3

Schedule 5

Schedule 5

Meeting Notes 7

Lab Meeting 1 9

Lab Meeting: 1 11

Lab Meeting 2 13

Lab Meeting: 2 15

Weekly Notes 17

GD Week 1 19

GD Weekly Notes: 1 21

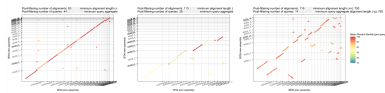
Shared/Overall To-do 21

Phylo-learning To-do 21

Bio-xAI To-do 21

About 23

Home



The shared electronic lab notebook of the Doing Com-Bio Lab.

Announcements

Something

Something else

Schedule

Weekly Schedule

Schedule

check often as things may change

W	TOPIC	Reading	Due
1	What is bioinformatics: what are computers and living cells good at?	Bioinformatics for Dummies (BFD) Ch. 1 & 2	Survey, Quiz 0
2	Algorithms & the Genetic Code	Think Python (TP) Ch. 1, 2, 3.4-3.7 & Molecular & Cell Biology (MCFD) Ch. 1, 6, 7	HW 1, Quiz 1
3	Modular Programming & Proteins	TP Ch. 5.1-5.7, 5.11, 6.1 & MCFD Ch. 16, 17	HW 2, Quiz 2
4	Loops & Sequencing	TP Ch. 6.1-6.1, 7.1-7.3, 7.7 & MCFD Ch. 19	HW 3, Quiz 3
5	Sequence Processing & Disease	BFD Ch. 7	HW 4, Quiz 4
6	Sequence Alignment	BFD Ch. 9, TP Ch. 8	HW 5, Quiz 5
7	Multiple Sequence Alignment	BFD Ch. 13	Project 1 Outline

W	TOPIC	Reading	Due
8	Polymerase Chain Reaction	BDF Ch. 13, 9 cont.	Project 1
9	Phylogenetics	BFD Ch. 13 cont.	Project 2 Check-in
10	Gene Prediction	BFD Ch. 5, pp. 145-153	Project 2

Meeting Notes

Lab Meeting 1

Lab Meeting: 1

June 15, 2026

1. lab meeting structure (first few weeks)
 - 10-15 mins each for project Q&A
 - maybe walk through notebooks from the previous week, if helpful
 - flash journal club
2. daily scrum - slack
 - home or lab
 - what you'll focus on
 - to know what Qs to expect or work together
 - to avoid doing the same thing
 - to get ideas from each other
 - approx hours
 - to synchronize if helpful
 - to know when to expect Qs or As
3. working notebooks vs. notebook notebooks
 - working colab for now - for reproducibility
 - data stored in the same google drive folder should be reproducible across accounts..
 - Union supposed to be getting a compute cluster
 - notebook notebooks can be edited in anything (I use VS code)
 - don't require reading in data?
4. github seqADAGE <https://github.com/georgiadoing/seqADAGE>

- fork
- clone
- make project folder within /Py/
- we'll start code review in a couple weeks
- [github tutorials](#)
 - maybe try the first..

5. data in google drive folder (for now)

- later will use Open Science Framework: <https://osf.io/>
- maybe [this one](#) but maybe a *new* one
- could create accounts (union email?)

6. Misc.

- my office key and keycard

Lab Meeting 2

Lab Meeting: 2

June 22, 2026

Weekly Notes

GD Week 1

GD Weekly Notes: 1

June 15, 2026

Shared/Overall To-do

- ☐ Implement tuner method in adage class
- ☐ E coli gene lists for enrichment analyses
- ☒ Get lab notebook setup
- ☐ Decide on repo structure
- ☐ Plan sampling excursion
- ☒ Key access to CAP/crochet
- ☒ Schedule lab meetings - update calendar
- ☒ Group meeting once a week
- ☐ 1-1 meetings also weekly or as-needed

Phylo-learning To-do

- ☒ Get DNA sequences into google drive
- ☐ Get P.a. DNA seq
- ☐ S aureus gene lists for enrichment analysis
- ☐ P aeruginosa gene lists for enrichment analysis

Bio-xAI To-do

- ☐ model to test SHAP tutorial on

☐ colaboration with Josh/Ara

About

References

